



**RAMAIAH**  
Institute of Technology

**Ramaiah Institute of Technology, Bengaluru-560054**

**Department of Mechanical Engineering**  
**(Autonomous Institute, Affiliated to VTU)**

**CIE Test-I**

**Sem/Section:** II Sem/E Sec

**Subject Code:** MELC28

**Maximum Marks:** 50

**Subject:** Computer Aided Engineering Drawing

**Duration:** 3 hours

**Instructions to Candidates:**

- Answer one full question from each unit.
- Draw solutions using pencil and scale to the dimensions.
- Answer Unit 1 questions in sketch work only, no computer solution is needed.
- For Unit-2 & Unit-3, submit both sketch work and computer solutions.

| Q.No. | UNIT-I                                                                                                                                                                                                                                                                                          | Mark      | COS        | BL        |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------------|-----------|
|       |                                                                                                                                                                                                                                                                                                 | <b>S</b>  |            |           |
| 1.    | A pentagonal lamina of edges 30 mm is resting on HP with one of its corners such that the edge opposite to this corner is 20mm above HP and makes an angle of $40^\circ$ with VP. Draw the top and front views of the lamina in this position. Determine the inclination of the lamina with HP. | <b>10</b> | <b>CO1</b> | <b>L6</b> |
| 2.    | A pentagonal prism of base side 25 mm, height 60 mm rests with its base on HP and two faces equally inclined to VP. Draw the development of the bottom portion of the prism if it is cut by a section plane bisecting the axis at $30^\circ$ to HP and perpendicular to VP.                     | <b>10</b> | <b>CO1</b> | <b>L6</b> |
|       | <b>UNIT-II</b>                                                                                                                                                                                                                                                                                  |           |            |           |
| 3.    | A hexagonal pyramid of base side 25 mm and 50 mm axis length rests on HP on one of its slant edges. Draw the projections of the pyramid when the axis of the top view is inclined to VP at $40^\circ$ .                                                                                         | <b>20</b> | <b>CO2</b> | <b>L6</b> |
| 4.    | A cylinder of base 50 mm diameter and axis 80 mm long lies on HP with its axis inclined at $45^\circ$ to HP and appears to be inclined at $30^\circ$ to VP. Draw the top and front views of the cone. <i>cylinder</i>                                                                           | <b>20</b> | <b>CO2</b> | <b>L6</b> |
|       | <b>UNIT-III</b>                                                                                                                                                                                                                                                                                 |           |            |           |
| 5.    | A cone with a base diameter of 40 mm and height of 60 mm rests centrally over a frustum of the hexagonal pyramid with a base side of 40 mm, top side of 35 mm, and height of 55 mm. Draw the isometric projection of the combined solid.                                                        | <b>20</b> | <b>CO3</b> | <b>L6</b> |
| 6.    | A regular pentagonal prism of base sides 30 mm and an axis of 60 mm is mounted centrally over a cylindrical block of 80 mm diameter and 25mm thick. Draw the isometric projection of the combined solids.                                                                                       | <b>20</b> | <b>CO3</b> | <b>L6</b> |